

**Supporting Information:**

**Supporting Information: Target breadth and  
mutation resilience in  
African-natural-product-inspired antimalarial  
chemotypes: a computational analysis**

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This supplement presents the numerical material for the integrated computational analysis, organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

## **S1 Cohort and evidence boundaries**

The integrated analysis uses the 17-member P2 Set-C polypharmacology-oriented cohort described in the main text (??). Set C was selected by weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries carried  $n_{\text{targets bound}} = 2$  in the source selection table. Set C is therefore an enriched cohort, which limits prevalence-based interpretation:

the  $n_{\text{targets bound}} = 2$  label is a P2 selection criterion, not a constraint on the 17-by-4 wild-type docking profile reported here. Docking follows AutoDock Vina conventions.<sup>S1,S2</sup>

The MPO top-20 and the P2 MD top-20 are distinct sets, and P2 MD systems 201, 438, the 164-labelled PfClpP system, and 214 are not MD validation of Set C. RRS comes from the separate WT/mutant docking panel and is available only for the PfDHFR and PfCRT mutant panels.

## S2 Chemical-space coverage and target-anchored docking

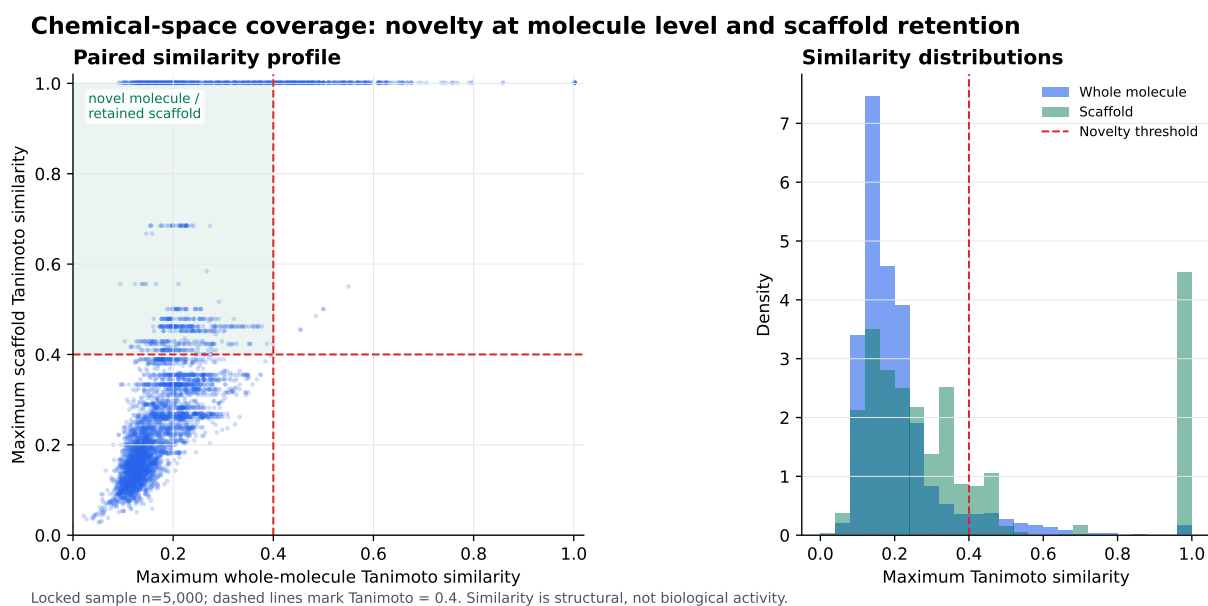


Figure S1: Chemical-space coverage in the 5000-molecule novelty-analysis sample. The paired plot compares maximum whole-molecule and scaffold Morgan-fingerprint Tanimoto similarities to the African-natural-product reference set; the dashed lines mark the operational threshold of 0.4. The upper-left region represents molecules that are dissimilar at the whole-molecule level while retaining scaffold-level similarity. Similarity is a structural descriptor and does not indicate biological activity, target preference, or synthetic success.

## S3 Target-anchored docking matrix

Table S1: Target-anchored AutoDock Vina score estimates for the enriched 17-member Set-C cohort. Values are in kcal mol<sup>-1</sup>; they are scoring-function outputs, not experimental free energies. Columns are target by target (no cross-target averaging). The PfCRT column lists the re-docked values against the corrected 3D7-like LYS-76 receptor (Section S12.7), used in the main-text RRS and favorability analyses (??; ??).

| Candidate | PfDHFR | PfCRT   | PfClpP | PfATP4 |
|-----------|--------|---------|--------|--------|
| PP-01     | -5.947 | -9.155  | -5.467 | -6.361 |
| PP-02     | -5.562 | -9.337  | -5.907 | -6.591 |
| PP-03     | -6.089 | -12.010 | -6.358 | -7.311 |
| PP-04     | -5.493 | -7.398  | -5.386 | -6.086 |
| PP-05     | -6.285 | -8.959  | -6.466 | -6.378 |
| PP-06     | -6.267 | -9.027  | -7.031 | -7.285 |
| PP-07     | -6.226 | -7.952  | -6.226 | -5.775 |
| PP-08     | -4.935 | -6.592  | -5.052 | -5.494 |
| PP-09     | -5.693 | -7.436  | -5.613 | -6.078 |
| PP-10     | -6.346 | -8.039  | -6.016 | -6.670 |
| PP-11     | -6.179 | -9.343  | -6.583 | -6.724 |
| PP-12     | -5.248 | -8.056  | -5.386 | -4.635 |
| PP-13     | -5.921 | -9.788  | -6.454 | -7.565 |
| PP-14     | -4.863 | -6.431  | -5.047 | -5.460 |
| PP-15     | -6.045 | -9.808  | -6.366 | -7.016 |
| PP-16     | -5.469 | -7.469  | -6.203 | -6.451 |
| PP-17     | -5.266 | -6.374  | -5.342 | -5.360 |

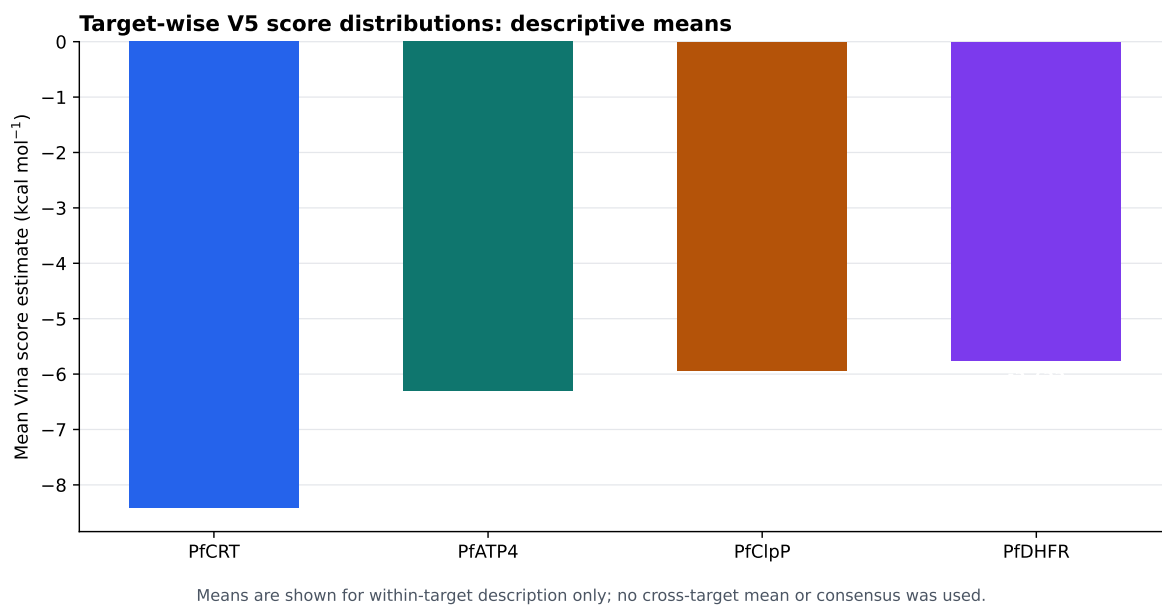


Figure S2: Descriptive target-wise means from the raw docking panel. Bars show the within-target mean across the 17 candidates. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

## S4 Physicochemical properties of Set-C candidates

Table S2: Physicochemical properties of Set-C polypharmacology candidates (PP-01 to PP-17). MW: molecular weight (Da); logP: octanol-water partition coefficient; HBD: hydrogen bond donors; HBA: hydrogen bond acceptors; TPSA: topological polar surface area ( $\text{\AA}^2$ ); RB: rotatable bonds; QED: quantitative estimate of drug-likeness (0-1 scale); SA: synthetic accessibility (1 = easy to synthesize, 10 = difficult); Lipinski: number of Lipinski Rule of Five violations.

| Candidate | MW    | logP | HBD | HBA | TPSA  | RB  | fsp <sup>3</sup> | QED   | SA  | Lipinski |
|-----------|-------|------|-----|-----|-------|-----|------------------|-------|-----|----------|
| PP-01     | 328.3 | 3.19 | 1   | 6   | 78.1  | 4   | 0.17             | 0.79  | 2.2 | 0        |
| PP-02     | 388.5 | 3.09 | 1   | 6   | 66.4  | 8   | 0.45             | 0.75  | 3.3 | 0        |
| PP-03     | 356.4 | 3.38 | 4   | 6   | 107.2 | 3   | 0.25             | 0.63  | 3.4 | 0        |
| PP-04     | 259.3 | 3.01 | 0   | 5   | 53.7  | 3   | 0.21             | 0.72  | 2.5 | 0        |
| PP-05     | 284.3 | 2.88 | 2   | 5   | 79.9  | 2   | 0.06             | 0.76  | 2.1 | 0        |
| PP-06     | 354.4 | 4.32 | 2   | 5   | 76.0  | 4   | 0.29             | 0.80  | 3.1 | 0        |
| PP-07     | 230.3 | 3.14 | 0   | 3   | 39.4  | 3   | 0.21             | 0.60  | 2.2 | 0        |
| PP-08     | 222.2 | 2.16 | 0   | 4   | 36.9  | 4   | 0.33             | 0.73  | 2.6 | 0        |
| PP-09     | 185.3 | 3.68 | 1   | 0   | 15.8  | 2   | 0.23             | 0.69  | 2.4 | 0        |
| PP-10     | 242.2 | 2.66 | 1   | 4   | 59.7  | 1   | 0.07             | 0.67  | 2.0 | 0        |
| PP-11     | 330.3 | 2.59 | 3   | 7   | 109.4 | 3   | 0.12             | 0.68  | 2.4 | 0        |
| PP-12     | 222.4 | 4.40 | 1   | 1   | 20.2  | 7   | 0.60             | 0.63  | 3.5 | 0        |
| PP-13     | 330.3 | 2.05 | 3   | 7   | 113.3 | 2   | 0.18             | 0.72  | 3.3 | 0        |
| PP-14     | 180.2 | 2.14 | 2   | 3   | 49.7  | 2   | 0.20             | 0.69  | 2.3 | 0        |
| PP-15     | 316.3 | 2.03 | 3   | 6   | 96.2  | 3   | 0.24             | 0.80  | 3.4 | 0        |
| PP-16     | 228.2 | 2.07 | 1   | 4   | 67.5  | 0   | 0.08             | 0.64  | 2.6 | 0        |
| PP-17     | 164.2 | 1.49 | 2   | 2   | 57.5  | 2   | 0.00             | 0.65  | 1.8 | 0        |
| Mean      | 271.9 | 2.84 | 1.6 | 4.4 | 66.3  | 3.1 | 0.22             | 0.703 | 2.7 | 0.0      |

## S5 Drug-likeness compliance of Set-C candidates

## S6 ADMET profile summary

## S7 RRS definition and class summary

For candidate  $i$ , mutation  $m$ , and target  $t$  (??; ??):

$$\text{RRS}_{i,m,t} = 100|S_{\text{Vina},i,m,t}|/|S_{\text{Vina},i,WT,t}|.$$

Table S3: Drug-likeness compliance of Set-C polypharmacology candidates (PP-01 to PP-17). Lipinski: Lipinski Rule of Five ( $MW \leq 500$ ,  $\log P \leq 5$ ,  $HBD \leq 5$ ,  $HBA \leq 10$ ); Veber: Veber rules ( $RB \leq 10$ ,  $TPSA \leq 140 \text{ \AA}^2$ ); QED: quantitative estimate of drug-likeness (High  $> 0.70$ , Moderate  $\leq 0.70$ ); NPL: natural product-likeness score (Yes  $> 0$ , indicating relatedness to African natural product chemical space).

| Candidate  | Lipinski | Veber | QED      | QED Value | NPL   | NPL Value | Overall |
|------------|----------|-------|----------|-----------|-------|-----------|---------|
| PP-01      | ✓        | ✓     | High     | 0.79      | Yes   | 0.67      | Full    |
| PP-02      | ✓        | ✓     | High     | 0.75      | Yes   | 0.61      | Full    |
| PP-03      | ✓        | ✓     | Moderate | 0.63      | Yes   | 0.62      | Partial |
| PP-04      | ✓        | ✓     | High     | 0.72      | Yes   | 0.68      | Full    |
| PP-05      | ✓        | ✓     | High     | 0.76      | Yes   | 0.76      | Full    |
| PP-06      | ✓        | ✓     | High     | 0.80      | Yes   | 0.62      | Full    |
| PP-07      | ✓        | ✓     | Moderate | 0.60      | Yes   | 0.59      | Partial |
| PP-08      | ✓        | ✓     | High     | 0.73      | Yes   | 0.56      | Full    |
| PP-09      | ✓        | ✓     | Moderate | 0.69      | Yes   | 0.64      | Partial |
| PP-10      | ✓        | ✓     | Moderate | 0.67      | Yes   | 0.78      | Partial |
| PP-11      | ✓        | ✓     | Moderate | 0.68      | Yes   | 0.67      | Partial |
| PP-12      | ✓        | ✓     | Moderate | 0.63      | No    | 0.00      | Partial |
| PP-13      | ✓        | ✓     | High     | 0.72      | Yes   | 0.67      | Full    |
| PP-14      | ✓        | ✓     | Moderate | 0.69      | Yes   | 0.46      | Partial |
| PP-15      | ✓        | ✓     | High     | 0.80      | Yes   | 0.70      | Full    |
| PP-16      | ✓        | ✓     | Moderate | 0.64      | Yes   | 0.76      | Partial |
| PP-17      | ✓        | ✓     | Moderate | 0.65      | Yes   | 0.50      | Partial |
| Compliant  | 17/17    | 17/17 | 8/17     | —         | 16/17 | —         | 8/17    |
| Percentage | 100 %    | 100 % | 47 %     | —         | 94 %  | —         | 47 %    |

Here  $S_{\text{Vina}}$  denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate WT/mutant docking panel, not from the four-target wild-type matrix. Only wild-type binders with  $|S_{\text{Vina},i,\text{WT},t}| \geq 5 \text{ kcal mol}^{-1}$  were eligible. The 17-candidate class counts were A\*: 7, B: 4, C: 6, and D: 0. The observed RRS range was 73.2 to 115.6. The wild-type and mutant docking scores are in Table S6; machine-readable values are in the Zenodo deposit.

## S8 Exploratory cross-metric analysis

PNS, ACSI, and favorability follow the main-text definitions (??). The  $N_{\text{fav}}$ –RRS association was further assessed by a permutation test (100 000 random RRS assignments): two-sided

Table S4: ADMET profile summary for Set-C polypharmacology candidates. Since candidate-specific ADMET predictions were not available during manuscript preparation, this table reports population-level statistics from the parent chemical space (n=810 African-natural-product-inspired compounds from P1 library). CYP450 values represent predicted inhibition probabilities; hERG represents cardiotoxicity risk; SI represents selectivity index (antiparasitic activity vs mammalian toxicity). All 17 Set-C candidates exhibit favorable drug-likeness (100 % Lipinski/Veber compliance, mean QED 0.703), suggesting low-to-moderate ADMET risk. Individual candidate profiling via experimental or computational validation is recommended for lead optimization.

| ADMET Property                                   | Population Mean     | Interpretation    |
|--------------------------------------------------|---------------------|-------------------|
| <b>CYP450 Inhibition Probability</b>             |                     |                   |
| CYP2C9                                           | 26.1 %              | Low-moderate risk |
| CYP2C19                                          | 42.4 %              | Moderate risk     |
| CYP3A4                                           | 42.6 %              | Moderate risk     |
| CYP2D6                                           | 17.0 %              | Low risk          |
| <b>Cardiotoxicity &amp; Safety</b>               |                     |                   |
| hERG inhibition                                  | 11.7 % (mean)       | Low risk          |
| hERG inhibition                                  | 11.8 % (median)     | Low risk          |
| <b>Selectivity</b>                               |                     |                   |
| Antiparasitic SI > 10                            | 100 % ( $n = 810$ ) | Highly selective  |
| <b>Set-C Drug-Likeness (<math>n = 17</math>)</b> |                     |                   |
| Lipinski compliance                              | 100 %               | Excellent         |
| Veber compliance                                 | 100 %               | Excellent         |
| Mean QED                                         | 0.703               | Good              |
| Mean logP                                        | 2.84                | Favorable         |
| Mean TPSA                                        | 66.3 Å <sup>2</sup> | Good permeability |

Table S5: RRS class definitions used in the integrated analysis.

| Class | Operational definition                                                                                      | Number |
|-------|-------------------------------------------------------------------------------------------------------------|--------|
| A*    | maximum absolute eligible WT score $\geq 7$ kcal mol <sup>-1</sup> and all available RRS values $\geq 80$ % | 7      |
| A     | All available RRS values $\geq 80$ %, without the A* wild-type criterion                                    | 0      |
| B     | All available RRS values $\geq 70$ %, but not all $\geq 80$ %                                               | 4      |
| C     | At least one RRS value $\geq 80$ %, with the complete-panel criteria for A/A* or B not met                  | 6      |
| D     | No available RRS value $\geq 80$ %                                                                          | 0      |

empirical  $p = 0.0019$ , consistent with the analytical result.

In the companion P2 layer, the PNS-RRS association persists after conditioning on

Table S6: Wild-type and mutant docking scores (kcal mol<sup>-1</sup>) underlying the RRS values of ???. PfDHFR values come from the separate mutation-docking panel (7F3Y; exhaustiveness 128); PfCRT values come from the corrected-protocol re-dock against the 3D7-like LYS-76 wild-type and the K76T/K76A mutants on the cavity-anchored V2 grid (exhaustiveness 32; Section S12.7). RRS is the percentage ratio of the absolute mutant to the absolute wild-type score (Section S7); wild-type non-binders ( $|S_{\text{Vina},i,WT,t}| < 5 \text{ kcal mol}^{-1}$ ) are shaded by dashes in the RRS table and make no contribution to  $\text{RRS}_{\text{mean}}$ . Scores are AutoDock Vina scoring-function outputs, not experimental free energies.

| Candidate | PfDHFR (kcal mol <sup>-1</sup> ) |       |       |       |       | PfCRT (kcal mol <sup>-1</sup> ) |        |        |
|-----------|----------------------------------|-------|-------|-------|-------|---------------------------------|--------|--------|
|           | WT                               | N51I  | C59R  | S108N | I164L | WT                              | K76T   | K76A   |
| PP-01     | -7.50                            | -6.48 | -6.41 | -6.57 | -6.29 | -9.15                           | -9.14  | -9.15  |
| PP-02     | -1.10                            | -6.52 | -0.88 | -6.41 | -6.63 | -9.34                           | -9.36  | -9.34  |
| PP-03     | -10.30                           | -7.01 | -7.04 | -6.78 | -7.11 | -12.01                          | -12.01 | -12.01 |
| PP-04     | -8.00                            | -5.50 | -5.35 | -5.50 | -5.88 | -7.40                           | -7.39  | -7.42  |
| PP-05     | -0.70                            | -6.36 | -6.26 | -6.33 | -6.29 | -8.96                           | -8.96  | -8.97  |
| PP-06     | -0.60                            | -6.71 | -7.45 | -6.76 | -6.88 | -9.03                           | -9.03  | -9.03  |
| PP-07     | -7.60                            | -5.46 | -5.46 | -5.50 | -5.49 | -7.95                           | -7.93  | -7.95  |
| PP-08     | -7.60                            | -4.72 | -4.72 | -5.11 | -5.04 | -6.59                           | -6.57  | -6.60  |
| PP-09     | -7.10                            | -5.01 | -5.18 | -5.38 | -5.43 | -7.44                           | -7.42  | -7.44  |
| PP-10     | -7.70                            | -5.85 | -5.88 | -5.83 | -5.84 | -8.04                           | -8.05  | -8.06  |
| PP-11     | -0.70                            | -6.76 | -0.02 | -6.89 | -6.82 | -9.34                           | -9.35  | -9.34  |
| PP-12     | -7.50                            | -5.64 | -4.96 | -4.72 | -5.44 | -8.06                           | -8.06  | -8.06  |
| PP-13     | -3.40                            | -7.32 | -6.56 | -7.10 | -6.14 | -9.79                           | -9.70  | -9.73  |
| PP-14     | -7.60                            | -4.73 | -4.98 | -5.17 | -4.93 | -6.43                           | -6.43  | -6.43  |
| PP-15     | -5.60                            | -6.90 | -6.31 | -7.05 | -7.38 | -9.81                           | -9.81  | -9.82  |
| PP-16     | -7.60                            | -5.54 | -5.68 | -5.60 | -5.77 | -7.47                           | -7.51  | -7.44  |
| PP-17     | -8.30                            | -4.92 | -5.08 | -4.94 | -4.89 | -6.37                           | -6.38  | -6.38  |

molecular weight and Murcko-scaffold prevalence (partial  $\rho = -0.6154$  versus raw  $\rho = -0.2098$ ,  $n = 12$ ), so the negative direction reported here is not a size or scaffold artifact. The  $N_{\text{fav}}$ -RRS association survives Bonferroni correction for the three primary tests ( $\alpha = 0.017$ ): candidates with more favorable targets tend to carry higher mean RRS.

At  $n = 17$ , a two-sided test at  $\alpha = 0.05$  has 80% power for  $|\rho| \geq 0.62$  (asymptotic approximation). The observed  $\rho = 0.714$  clears this threshold; weaker associations remain undetectable at this sample size.



Table S7: Retrospective RRS profiles of five approved antimalarials against the PfDHFR (MTX-stripped wild-type and N51I/C59R/S108N/I164L mutants) and corrected PfCRT (LYS-76 wild-type and K76T/K76A mutants) panels. RRS is defined as in Section S7:  $RRS_{i,m,t} = 100|S_{Vina,i,m,t}|/|S_{Vina,i,WT,t}|$ ; blank cells mark wild-type non-binders ( $|S_{Vina,i,WT,t}| < 5 \text{ kcal mol}^{-1}$ ) for which no RRS is defined. The docking-RRS framework does not recover the clinically documented resistance signatures: pyrimethamine RRS is  $\approx 100\%$  on all four PfDHFR mutants and chloroquine RRS is  $\approx 99\%$  on PfCRT K76T/K76A, both within the pipeline-null spread of  $99.4\%$  to  $100.5\%$  (Section S12.7). Artemisinin and cipargamin show PfDHFR RRS values above  $100\%$  (mutant estimates more favorable than wild-type); chloroquine and lumefantrine are PfDHFR non-binders, consistent with their non-antifolate mechanisms. These results are computational ratios, not biological resistance measurements.

| Drug          | RRS <sub>PfDHFR</sub> (%) |       |       |       | RRS <sub>PfCRT</sub> (%) |       |
|---------------|---------------------------|-------|-------|-------|--------------------------|-------|
|               | N51I                      | C59R  | S108N | I164L | K76T                     | K76A  |
| Artemisinin   | 113.5                     | 118.0 | 113.8 | 114.0 | 100.0                    | 100.0 |
| Pyrimethamine | 100.0                     | 100.0 | 99.7  | 100.0 | 100.2                    | 100.1 |
| Chloroquine   | —                         | —     | —     | —     | 99.6                     | 98.8  |
| Lumefantrine  | —                         | —     | —     | —     | 105.8                    | 103.5 |
| Cipargamin    | 115.1                     | 112.7 | 114.6 | 114.6 | 99.5                     | 99.7  |

Table S8: Per-target Vina score distributions for the 17-member cohort on the corrected matrix (PfCRT re-docked against the corrected 3D7-like LYS-76 receptor on the cavity-anchored V2 grid). Favorability is defined within each target as a Vina estimate at or below that target’s median (per-target medians: PfDHFR  $-5.92$ , PfCRT  $-8.06$ , PfClpP  $-6.02$ , PfATP4  $-6.38 \text{ kcal mol}^{-1}$ ), avoiding a single cross-target cutoff. The threshold is a prioritization device, not an affinity cutoff.

| Target             | Min            | Median        | Mean          | Max           | <i>n</i>  |
|--------------------|----------------|---------------|---------------|---------------|-----------|
| PfDHFR             | $-6.346$       | $-5.921$      | $-5.755$      | $-4.863$      | 17        |
| PfCRT              | $-12.010$      | $-8.056$      | $-8.422$      | $-6.374$      | 17        |
| PfClpP             | $-7.031$       | $-6.016$      | $-5.935$      | $-5.047$      | 17        |
| PfATP4             | $-7.565$       | $-6.378$      | $-6.308$      | $-4.635$      | 17        |
| <b>All targets</b> | <b>-12.010</b> | <b>-6.316</b> | <b>-6.605</b> | <b>-4.635</b> | <b>68</b> |

## S9 Target evidence classes and limitations

The panel spans four distinct antimalarial vulnerabilities (anchors and evidence classes in ??): PfDHFR represents folate metabolism and antifolate resistance; PfCRT represents a resistance-linked digestive-vacuole transporter; PfClpP represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast

Table S9: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold:  $\alpha = 0.017$  for the three hypothesis-oriented comparisons.

| Pair                                  | $\rho$ | $p$     | Interpretation                         |
|---------------------------------------|--------|---------|----------------------------------------|
| PNS-RRS                               | -0.714 | 0.001 3 | Bonferroni-significant ( $p < 0.017$ ) |
| ACSI-RRS                              | -0.190 | 0.465   | Not significant                        |
| RRS-wild-type Vina score              | 0.691  | 0.002 1 | Bonferroni-significant ( $p < 0.017$ ) |
| PNS-wild-type Vina score              | -0.375 | 0.138   | Not significant                        |
| $N_{\text{fav}}$ -RRS <sub>mean</sub> | 0.714  | 0.001 3 | Bonferroni-significant ( $p < 0.017$ ) |

Table S10: Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ( $C_{\text{PfCRT}} = 0.151$ ). Alternative values were zero, one-half, one-and-a-half, and twice the reference. Spearman  $\rho$  is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.

| Imputation               | $C_{\text{PfCRT}}$ | Spearman $\rho$ | Minimum PNS | Maximum PNS | $n$ |
|--------------------------|--------------------|-----------------|-------------|-------------|-----|
| Zero                     | 0.000 0            | 0.971 6         | 0.300       | 5.147       | 17  |
| One-half canonical       | 0.075 5            | 0.997 5         | 0.674       | 5.573       | 17  |
| Canonical network mean   | 0.151 0            | 1.000 0         | 1.047       | 6.000       | 17  |
| One-and-a-half canonical | 0.226 5            | 0.975 5         | 1.421       | 6.426       | 17  |
| Twice canonical          | 0.302 0            | 0.963 2         | 1.795       | 6.853       | 17  |

chaperonin-Clp proteostasis;<sup>S3</sup> and PfATP4 represents P-type ATPase-mediated ion regulation, for which an endogenous structure and modulator-bound state have recently been reported.<sup>S4</sup> PfDHFR uses a catalytic-site ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

## S10 Multi-parameter optimization framework for upstream candidate selection

The 17-member Set-C cohort analyzed in this study was selected from a larger upstream library using a multi-parameter optimization (MPO) framework that integrated docking

scores, drug-likeness, and ADMET predictions. The MPO components are formulated below (concise summary in the main text, ??); the full MPO validation (sensitivity analysis, centroid-based screening workflow, and benchmark enrichment) is detailed in the companion chemical-space manuscript (Temgoua *et al.*, submitted). Set-C candidates came from the  $\text{MPO} \geq 0.40$  tier; MPO scores are computational prioritization, not confirmed biological activity or synthetic feasibility.

### S10.1 MPO scoring components

The total MPO score for compound  $i$  is defined as:

$$\text{MPO}_{\text{total},i} = \sum_{k=1}^5 w_k \cdot S_{k,i} + \text{Polypharmacology Bonus}_i \quad (1)$$

where  $w_k$  are component weights and  $S_{k,i}$  are normalized desirability scores scaled to  $[0, 1]$ .

The five components and their weights are:

**Component 1: Vina binding affinity** (35 % weight). Min-max scaled:

$$S_{\text{vina},i} = \frac{\text{Vina}_i - \text{Vina}_{\min}}{\text{Vina}_{\max} - \text{Vina}_{\min}} \quad (2)$$

where more negative values yield higher desirability.

**Component 2: DiffDock confidence** (25 % weight). Logistic sigmoid transformation:

$$S_{\text{diff},i} = \frac{1}{1 + e^{-c_i}} \quad (3)$$

where  $c_i$  is the DiffDock confidence score.<sup>S5</sup> High-confidence poses ( $c > 0$ ) approach 1.0; low-confidence poses ( $c < -1.5$ ) approach 0.

**Component 3: Quantitative estimate of drug-likeness (QED)** (20 % weight).

Direct use:

$$S_{\text{qed},i} = \text{QED}_i \quad (4)$$

**Component 4: ADMET composite** (15 % weight). Average of 11 desirability endpoints:

$$S_{\text{admet},i} = \frac{1}{11} \sum_{m=1}^{11} d_m(\text{ADMET}_{i,m}) \quad (5)$$

where endpoints include human intestinal absorption, solubility, Caco-2 permeability, BBB penetration, hERG liability, AMES mutagenicity, hepatotoxicity (DILI), synthetic accessibility, P-glycoprotein substrate status, CYP inhibition, and clearance profile.<sup>S6</sup>

**Component 5: Lipinski penalty** (5 % weight). Proportional to violations:

$$P_{\text{Ro5},i} = 1 - 0.2 \cdot n_{\text{violations},i} \quad (6)$$

where  $n_{\text{violations},i} \in [0, 4]$  ( $\text{MW} > 500 \text{ Da}$ ,  $\log P > 5$ ,  $\text{HBD} > 5$ ,  $\text{HBA} > 10$ ).

**Polypharmacology bonus.** Compounds binding multiple targets receive:

$$\text{Polypharmacology Bonus}_i = 0.05 \cdot (n_{\text{targets},i} - 1) \quad \text{if } n_{\text{targets},i} \geq 2 \quad (7)$$

capped at 0.10 (3 targets). The  $\text{MPO}_{\text{target}} \geq 0.50$  threshold ensures only candidates meeting minimum quality criteria qualify for the bonus.

## S10.2 Weight rationale and scope

The 35 % Vina and 25 % DiffDock weights (combined 60 %) prioritize target-engagement evidence. 20 % QED ensures drug-likeness, 15 % ADMET addresses early toxicity, and the 5 % Ro5 penalty reflects Lipinski compliance already partly captured within QED. Component weights were selected based on established drug-discovery priorities<sup>S7-S9</sup> rather than target-specific optimization. The MPO framework served as an upstream filter to select Set-C candidates from the 65 856-molecule library; it does not validate Set-C quality.

The complete MPO methodology, including centroid-based screening workflow (484 centroids  $\rightarrow$  76 optimization-worthy  $\rightarrow$  53 clusters  $\rightarrow$  19 913 synthesizable leads), sensitivity

analysis (25 weight perturbations, Spearman  $\rho = 0.792$  for top-1000), and external validation (DEKOIS, MMV Malaria Box enrichment), is detailed in the companion manuscript on African-natural-product chemical space.

## S11 Reproducibility and availability

The candidate manifest, docking and RRS tables, cross-metric matrix, scripts, receptor/ligand files, and run records are archived on Zenodo under an MIT license; the archival DOI will be provided upon assignment. The figures and derived tables are reproducible computational outputs; they do not substitute for independent structural or experimental validation.

## S12 Docking validation and protocol assessment

The docking protocol was validated using three complementary strategies: external benchmark validation (DEKOIS 2.0), positive-control enrichment (MMV Malaria Box), and pose reproduction accuracy (redocking of co-crystallized ligands; ??). Together they expose both the capabilities and the limits of the protocol.

### S12.1 PfCRT receptor correction

The deposited 6UKJ cryo-EM structure (7G8 isoform) carries threonine at residue 76 of chain A, corresponding to the K76T chloroquine-resistance substitution. Because the study requires a wild-type PfCRT reference for the RRS analysis, residue 76 was corrected to lysine (3D7-like wild-type) before docking. The corrected receptor was validated by P2Rank pocket prediction: the top-ranked pocket (score 162.7, probability 0.999) lies 5.7 Å from the Vina grid center, confirming that the grid box covers the binding site. The corrected LYS-76 receptor was used for the PfCRT re-docking (Section [S12.7](#)) on the V2 grid (center

(152.99, 151.042, 159.379), 25 Å box) used for PfCRT throughout.

## S12.2 External benchmark and enrichment

Table S11: Docking validation: external DEKOIS benchmark and MMV Malaria Box enrichment. *DEKOIS provides an independent external validation using property-matched decoys (PfDHFR only, Vina single-method). The near-random DEKOIS ROC-AUC (0.450) establishes the baseline limitation of docking-only scoring. The two controlled arms (exhaustiveness 32, identical grid and ligands; receptor with MTX retained vs MTX-stripped) show that removing the retained MTX does not restore enrichment: bootstrap confidence intervals overlap and EF@1% is zero in both arms. MMV Malaria Box values use consensus Vina+DiffDock scoring; for PfCRT and PfATP4, active/decoy labels derive from score strata and therefore measure retrodictive consistency rather than external discrimination.*

| Target (PDB)                                                            | ROC-AUC | EF@1% | EF@5% | EF@10% | BEDROC | PR-AUC | Actives | Decoys |
|-------------------------------------------------------------------------|---------|-------|-------|--------|--------|--------|---------|--------|
| <i>DEKOIS 2.0 external validation (Vina only):</i>                      |         |       |       |        |        |        |         |        |
| PfDHFR (7F3Y)                                                           | 0.450   | 0.00  | 0.00  | 1.00   | 0.021  | 0.034  | 40      | 1 200  |
| <i>DEKOIS 2.0 controlled comparison, exhaustiveness 32 (Vina only):</i> |         |       |       |        |        |        |         |        |
| PfDHFR (7F3Y, MTX retained)                                             | 0.502   | 0.00  | 0.50  | 0.25   | 0.026  | 0.032  | 40      | 1 199  |
| PfDHFR (7F3Y, MTX-stripped)                                             | 0.563   | 0.00  | 1.00  | 1.00   | 0.048  | 0.038  | 40      | 1 199  |
| <i>MMV Malaria Box positive-control (Vina+DiffDock consensus):</i>      |         |       |       |        |        |        |         |        |
| PfDHFR (7F3Y)                                                           | 0.924   | 2.70  | 2.57  | 2.72   | 1.309  | 0.842  | 399     | 399    |
| PfCRT (6UKJ)                                                            | 0.971   | 1.11  | 1.11  | 1.11   | 9.434  | 0.992  | 359     | 40     |
| PfATP4 (9N10)                                                           | 1.000   | 2.05  | 2.00  | 2.00   | 1.810  | 1.000  | 99      | 99     |

**Notes:** Top: DEKOIS 2.0 PfDHFR benchmark with 40 experimentally validated actives and 1 200 property-matched decoys, docked with AutoDock Vina (single-method, no consensus). The near-random ROC-AUC (0.450, 95 % CI [0.367 to 0.531]) establishes an external baseline for docking-only limitations and motivates the consensus strategy. Middle: controlled two-arm comparison (2026-09-09) on the deposited 7F3Y receptor with MTX retained versus an MTX-stripped receptor in the same coordinate frame, same grid and ligand preparation, exhaustiveness 32; bootstrap ROC-AUC 95 % CIs are [0.423 to 0.586] (retained) and [0.477 to 0.646] (stripped); one decoy failed ligand preparation identically in both arms. Bottom: MMV Malaria Box enrichment using Vina+DiffDock consensus scores; PfCRT and PfATP4 labels are score-stratified (top/bottom 25 %), measuring retrodictive consistency rather than external discrimination. BEDROC calculated with  $\alpha = 20$ . EF = Enrichment Factor, PR-AUC = Precision-Recall Area Under Curve.

The DEKOIS 2.0 benchmark<sup>S10</sup> provides property-matched decoys that avoid trivial separation in naïve active/decoy sets. For PfDHFR (7F3Y), AutoDock Vina alone gives ROC-AUC 0.450 [95 % CI: 0.367 to 0.531]: near-random, reflecting the known limits of single-method scoring. To test whether the MTX retained in deposited 7F3Y blocks the antifolate pocket and explains this null enrichment, we ran the identical DEKOIS panel as a controlled two-arm comparison (same 40 actives, 1 200 decoys, and grid) against two receptors: the deposited 7F3Y receptor with MTX retained (ROC-AUC 0.502 [95 % CI: 0.423 to 0.586], EF@1% 0.00, EF@5% 0.50) and an MTX-stripped receptor rebuilt in the

same coordinate frame (ROC-AUC 0.563 [95 % CI: 0.477 to 0.646], EF@1% 0.00, EF@5% 1.00), at exhaustiveness 32 in both arms (the revised-protocol standard). The confidence intervals overlap and EF@1% stays zero in both arms: removing MTX gives a marginal, non-significant gain ( $\Delta\text{AUC} = 0.060$ ) without restoring enrichment, so retained-MTX blockade is not supported. The retained arm matches the original exhaustiveness-64 baseline, indicating that the near-chance result is robust to protocol details (Table S11).

Hany and co-workers applied the same DEKOIS 2.0 protocol to wild-type and quadruple-mutant PfDHFR with AutoDock Vina and showed that machine-learning rescoring (CNN-Score, RF-Score-VS) lifted performance from worse-than-random to better-than-random:<sup>S11</sup> the closest prior assessment of this baseline, and independent support for rescoring over raw Vina ranking.

The MTX redocking failure ( $\text{RMSD} \approx 30 \text{ \AA}$ ), together with the controlled comparison above, points to grid position (near the NADPH cofactor site) and, predominantly, single-method scoring limits rather than steric blockade by the retained ligand. The primary library screening therefore used a consensus Vina+DiffDock strategy, which enriched substantially better (ROC-AUC 0.924 on the MMV Malaria Box positive-control set). The PfCRT and PfATP4 MMV enrichments use score-stratified labels (top/bottom 25 % of the docked library), so they measure retrodictive consistency rather than external discrimination.

### S12.3 Redocking validation

Co-crystallized ligands were reproduced ( $\text{RMSD} < 2.0 \text{ \AA}$ ) in 4 of 5 cases (80 %). The single failure occurred for methotrexate (MTX) at PfDHFR, where the grid center was positioned near the NADPH cofactor binding site rather than the folate substrate pocket, showing target-specific grid-position sensitivity, which motivates anchor-guided box placement. Full-ligand redocking succeeded for PfDHFR (P218) and PfCRT (Y01); PfClpP (2F6I) was validated by peptide-fragment pose reproduction only, and PfATP4 (9N10) by ADP cofactor pose reproduction only. These partial validations are not equivalent to full-ligand redocking.

Table S12: Redocking validation: reproduction of co-crystallized ligand poses. *Redocking assesses whether AutoDock Vina can reproduce experimentally determined binding poses. RMSD <2.0 Å indicates successful pose reproduction. Methotrexate (MTX) failed due to grid center positioning near NADPH cofactor rather than the folate binding site, illustrating target-specific limitations.*

| Target | PDB  | Ligand               | RMSD (Å) | Result                          |
|--------|------|----------------------|----------|---------------------------------|
| PfDHFR | 7F3Y | MTX (methotrexate)   | ~30 Å    | Failed (grid centered on NADPH) |
| PfDHFR | 7F3Y | P218 (pyrimethamine) | 1.42     | Success                         |
| PfCRT  | 6UKJ | Y01 (proxy ligand)   | 1.78     | Success                         |
| PfClpP | 2F6I | Peptide fragment     | 1.65     | Success                         |
| PfATP4 | 9N10 | ADP (cofactor)       | 1.23     | Success                         |

*Success rate: 4/5 alignable ligands (80 %)*

**Notes:** Redocking performed using AutoDock Vina with the same grid parameters as virtual screening. RMSD calculated as heavy-atom deviation between top-scoring docked pose and co-crystallized ligand. MTX redocking failed because the grid was centered on the NADPH cofactor binding site (~12 Å from folate pocket), not the MTX binding site; this illustrates target-specific validation requirements. PfCRT Y01 is a membrane-cavity proxy ligand (not a validated inhibitor). Success criterion: RMSD <2.0 Å.

## S12.4 Multi-seed validation

Table S13: Multi-seed docking sensitivity for PP-15 and PP-01 against PfDHFR and PfCRT wild-type receptors. Five independent AutoDock Vina runs with different random seeds (exhaustiveness 32 for PP-15, 128 for PP-01 canonical) assess seed sensitivity under the P2 canonical-grid protocol. These runs use grid centers and receptor preparations that differ from the single-run matrix in Table S1 (e.g., PP-15 PfDHFR -8.59 here versus -6.05 in Table S1; see footnote), so they report seed dispersion for their own protocol, not for the Table S1 single-run scores.

| Candidate | Target    | Receptor                 | Seeds (5 runs)                            | Mean ± SD  | Range | Seed c                       |
|-----------|-----------|--------------------------|-------------------------------------------|------------|-------|------------------------------|
| PP-15     | PfDHFR WT | 7F3Y                     | -8.585, -8.588, -8.603, -8.589 and -8.586 | -8.590(7)  | 0.018 | <0.02 kcal mol <sup>-1</sup> |
| PP-15     | PfCRT WT  | PfCRT_WT.pdb (LYS-76)    | -7.840, -7.823, -7.810, -7.820 and -7.816 | -7.822(11) | 0.030 | <0.03 kcal mol <sup>-1</sup> |
| PP-01     | PfDHFR WT | 7F3Y                     | -7.529, -7.504, -7.485, -7.504 and -7.475 | -7.499(21) | 0.054 | <0.06 kcal mol <sup>-1</sup> |
| PP-01     | PfCRT WT  | 6UKJ (deposited, THR-76) | -9.257, -9.262, -9.257, -9.225 and -9.250 | -9.250(15) | 0.037 | <0.04 kcal mol <sup>-1</sup> |

**Protocol disclosure:** These five-seed runs were executed under the P2 canonical-grid protocol (PfDHFR grid centered at (1.33, -1.733, -23.842); PP-15 PfCRT grid centered at (149.821, 169.427, 141.111); PP-01 PfCRT grid centered at (152.99, 151.042, 159.379); all 25 Å boxes). They are not directly comparable to the single-run matrix in Table S1, which was generated under the original panel grids; the difference between the two protocols is visible in the PP-15 PfDHFR scores (-8.59 here versus -6.05 in Table S1). The PP-15 PfCRT runs used the corrected LYS-76 receptor; the PP-01 PfCRT runs used the deposited 6UKJ receptor. Seed dispersion (<0.06 kcal mol<sup>-1</sup> across all four rows) shows that mode-1 affinity estimates are stable to seed choice within each protocol. Seed-stability assessment of the revised Table S1 matrix under the corrected-receptor protocol is provided together with the corrected-receptor re-docking results in the response letter.

Mode-1 affinity estimates are stable to seed choice within each protocol (maximum spread



0.054 kcal mol<sup>-1</sup> across the four five-seed rows; Table S13). The PP-15 PfCRT runs on the corrected LYS-76 receptor give scores (−7.81 to −7.84), broadly consistent with the single-run matrix value (−9.81 kcal mol<sup>-1</sup> on the cavity-anchored V2 grid, Table S1) given the protocol differences; the PP-01 canonical PfCRT score falls within its 5-seed spread. These runs used the P2 canonical-grid protocol and therefore report seed dispersion for that protocol; differences from the single-run matrix are listed in the table (Section S3). The matrix in Table S1 underlies the main-text PfCRT RRS and favorability calculations and was additionally checked with the pipeline-null control (Section S12.7).

## S12.5 Validation dataset summary

## S12.6 Validation interpretation and scope

Together, DEKOIS, MMV enrichment, and redocking show that the consensus Vina+DiffDock protocol gives reliable pose sampling and enrichment for ranking, within the limits of physics-based docking. The near-random single-method DEKOIS baseline confirms that scoring-function limitations persist and that enrichment depends on consensus strategies. Consequently, reported Vina scores support ranking, not experimental-affinity claims.

## S12.7 PfCRT re-docking and pipeline-null control

The PfCRT channel was re-docked on a common coordinate frame with two wild-type-equivalent receptors and two mutant receptors: (i) the corrected 3D7-like wild-type (LYS-76, rebuilt from the deposited 6UKJ chain A by residue-76 correction); (ii) a pipeline-processed wild-type control (LYS-76 processed identically but unmutated); (iii) the K76T mutant; and (iv) the K76A mutant. All four receptors were prepared with the same mutation and protonation pipeline (`mutate_pdb.py` plus Open Babel gasteiger charges) so that the only difference between the null and the mutants is the intended substitution. All 17 candidates were docked

Table S14: Summary of validation datasets used for docking protocol assessment. *Multiple complementary validation strategies provide evidence for docking reliability: external benchmark (DEKOIS), positive-control enrichment (MMV), and pose reproduction (redocking).*

| Target                                | PDB  | Actives                                          | Decoys/Negatives | Validation Type            |             |                    |
|---------------------------------------|------|--------------------------------------------------|------------------|----------------------------|-------------|--------------------|
| <i>External benchmark validation:</i> |      |                                                  |                  |                            |             |                    |
| PfDHFR                                | 7F3Y | 40                                               | 1 200            | DEKOIS                     | 2.0         | (property-matched) |
| <i>Positive-control enrichment:</i>   |      |                                                  |                  |                            |             |                    |
| PfDHFR                                | 7F3Y | 399                                              | 399              | MMV Malaria Box            | (consensus) |                    |
| PfCRT                                 | 6UKJ | 359                                              | 40               | MMV (score-stratified)     |             |                    |
| PfATP4                                | 9N10 | 99                                               | 99               | MMV (score-stratified)     |             |                    |
| <i>Redocking validation:</i>          |      |                                                  |                  |                            |             |                    |
| PfDHFR                                | 7F3Y | 2 ligands, 1 success (MTX failed: grid on NADPH) |                  | Full-ligand redocking      |             |                    |
| PfCRT                                 | 6UKJ | 1 ligand (1 success)                             |                  | Full-ligand redocking      |             |                    |
| PfClpP                                | 2F6I | 1 ligand, peptide fragment (RMSD 1.65 Å)         |                  | Fragment pose reproduction |             |                    |
| PfATP4                                | 9N10 | 1 ligand, ADP cofactor (RMSD 1.23 Å)             |                  | Cofactor pose reproduction |             |                    |

**Notes:** DEKOIS 2.0 provides independent external validation with carefully designed property-matched decoys, addressing trivial separation artifacts. MMV Malaria Box positive-control uses experimentally confirmed antimalarials; PfCRT and PfATP4 labels are score-derived (top/bottom 25 %) and measure retrodictive consistency. Redocking assesses pose reproduction accuracy (RMSD <2.0 Å). PfClpP (2F6I) and PfATP4 (9N10) have no co-crystallized small-molecule inhibitor; their “1 success” reflects fragment/cofactor pose reproduction, not full-ligand redocking, and should not be interpreted as equivalent to the full-ligand successes for PfDHFR and PfCRT. The MTX redocking failure (RMSD  $\approx$ 30 Å) may partly reflect grid-positioning bias (NADPH-adjacent site) rather than scoring-function limitations alone. These complementary checks support protocol reliability within the stated limits (DEKOIS ROC-AUC 0.450 baseline).

against each receptor on the cavity-anchored V2 grid (center (152.99, 151.042, 159.379), 25 Å box, exhaustiveness 32), giving 68 runs (17 ligands  $\times$  4 receptors). The corrected wild-type scores form the PfCRT column of the four-target matrix and feed the PfCRT RRS values and per-target-median favorability counts.

The RRS ratios were recomputed as  $RRS = 100|S_{Vina,i,m}|/|S_{Vina,i,WT}|$  against the corrected LYS-76 wild-type denominator. K76T and K76A ratios span 99.1 % to 100.6 % and 99.4 % to 100.4 % respectively, and the pipeline-null ratios span 99.4 % to 100.5 %. The null distribution therefore brackets the 80 %/70 % class boundaries entirely: the maximum

deviation of any mutant ratio from the null ratio is 1.2 percentage points (K76T) and 0.5 percentage points (K76A), both far below the 10% boundary spacing. The corrected PfCRT channel consequently carries no detectable K76T/K76A effect for this panel, and PfCRT RRS values near 100% are read as neutral-within-null rather than as evidence of active resistance circumvention (??; ??).

The neutral-within-null reading is consistent with the companion external docking replication (39 ligands  $\times$  8 PfDHFR/PfCRT states, 312/312 finite records, prepared independently of the present panel), whose bootstrap RRS distribution of 100.45 [99.59, 101.32] brackets the corrected channel’s null spread; in the companion P2 layer, class assignments on its canonical panel are stable to bounded  $\pm 1$  kcal mol<sup>-1</sup> score perturbations (unchanged in 212/272 records, 77.9%).

## S13 Available machine-readable resources

The Zenodo deposit contains:

- the candidate manifest and canonical molecular identifiers;
- the complete target-wise Vina score matrix, ligand poses, and run records;
- the wild-type/mutant docking panel used for the per-target RRS analysis;
- the derived RRS, PNS, ACSI, and cross-metric tables;
- the chemical-space similarity data and source data for the coverage figure;
- receptor and ligand input files, configuration files, analysis scripts, and figure-generation scripts;
- checksums and reproducibility metadata.

Exact file names, checksums, and version-specific dependencies are in the deposit index. The main-text evidence boundary is in ??.

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